

ESP32 Fingerprint Attendance System

Pre-Handover Setup Guide

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System	ESP32 Fingerprint Attendance — D and D Academy
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1. Google Sheets Setup

1.1 Create the School Spreadsheet

The system uses a single Google Spreadsheet to store all attendance and enrollment data. There is no central routing spreadsheet — the Apps Script writes directly to one spreadsheet.

1. Go to sheets.google.com and click **Blank spreadsheet** inside the Google account that will own the Apps Script.
2. Rename it clearly — e.g. *D and D Academy — Staff Attendance*.
3. Copy the Spreadsheet ID from the URL bar (the long string between /d/ and /edit). You will paste this into the Apps Script config as **SCHOOL_SPREADSHEET_ID**.

1.2 Attendance and Enrollment Tabs

Both the Attendance tab and the Enrollment tab are created automatically by the Apps Script — no manual tab creation is required. The Enrollment tab is created the first time the device polls for enrollment jobs (within 10 seconds of boot, if WiFi is connected).

[WARNING]

If enrollment commands are silently failing, check that the Apps Script is deployed and reachable. The Enrollment tab cannot be created if the script returns an error.

2. Apps Script Deployment

2.1 Open the Script Editor

1. In the school spreadsheet, click **Extensions** → **Apps Script**.
2. Delete any existing code in the editor window.
3. Paste the full contents of **backend/Staff_Attendance_Apps_Script.js**.

2.2 Set the Configuration Constants

Locate the **/* CONFIG */** section at the top of the script and fill in the following values:

Constant	Value for this deployment
SCHOOL_SPREADSHEET_ID	The spreadsheet ID copied in step 1.1
SCHOOL_NAME	"D and D Academy"
SHIFT_START_HOUR / SHIFT_START_MIN	7 / 30 (school day starts 7:30 AM)
SHIFT_END_HOUR / SHIFT_END_MIN	15 / 30 (school day ends 3:30 PM)
LATE_GRACE_MINUTES	10 (10-minute tolerance before marking Late)
EARLY_GRACE_MINUTES	10 (10-minute tolerance before marking Left-Early)
adminEmails	Array of email addresses to receive the monthly report
SCRIPT_AUTH_KEY	A secret string — e.g. "dda-auth-2025". Set the same value in the firmware. Leave " to disable.

2.3 Deploy as a Web App

1. Click **Deploy** → **New deployment**.
2. Click the gear icon next to "Select type" and choose **Web app**.
3. Set **Execute as: Me**.
4. Set **Who has access: Anyone**.
5. Click **Deploy**. Authorise the script when prompted — grant all requested permissions.
6. Copy the Web app URL that appears (ends in /exec). This is the **SCRIPT_URL** for the firmware.

[NOTE]

Every time you edit the script and want the changes to go live, you must create a New deployment — updating an existing deployment does NOT update the live app. After each new deployment, update SCRIPT_URL in the firmware and re-flash the device.

2.4 Set Up Time-Driven Triggers

Two functions must run automatically on a schedule.

1. Click the clock icon (Triggers) in the left sidebar of the Apps Script editor.
2. Click **+ Add Trigger** (bottom right).
3. Configure the nightly job trigger: Function: **runNightlyJob**, Event source: Time-driven, Type: Day timer, Time: **10pm–11pm**.

[NOTE]

The school day ends at 3:30 PM. Running the nightly job at 10pm–11pm gives ample time after the last staff member has left, ensuring no valid check-outs are missed.

4. Click **Save**. Add a second trigger: Function: **sendMonthlySummary**, Event source: Time-driven, Type: Month timer, Day: **1**, Time: **1am–2am**.

[NOTE]

The monthly summary covers the previous **calendar month** (1st through last day of the previous month). Running on the 1st at 1am–2am ensures the nightly job has already closed out the last day of the previous month. When the monthly report runs, the previous month's attendance rows are archived to a dedicated tab inside the spreadsheet and then deleted from the Attendance sheet; the full data set is also preserved in the XLSX email attachment sent to admins.

3. Firmware Configuration and Flash

3.1 Required Libraries (Arduino IDE)

Library	Install via
Adafruit Fingerprint Sensor Library	Arduino Library Manager
RTCLib (Adafruit)	Arduino Library Manager
ESP32 board support package	Board Manager — esp32 by Espressif

3.2 Edit the Firmware Config Block

Open **firmware/StaffAttendance_School.ino** in the Arduino IDE and update the ***/* CONFIG */*** section at the top:

Constant	Value for this deployment
SCRIPT_URL	The Web App URL from step 2.3
SCRIPT_AUTH	Same value as SCRIPT_AUTH_KEY in the script. Leave "" if disabled.
SCHOOL_ID	"DDA"
SCHOOL_NAME	"D and D Academy"
R503_RX_PIN / R503_TX_PIN	16 / 17 (default — match your wiring)

[NOTE]

WiFi credentials are NOT set in the firmware. They are entered via the captive portal on first boot and stored in SPIFFS.

3.3 Format SPIFFS and Upload Filesystem

1. In the Arduino IDE, go to **Tools** → **ESP32 Sketch Data Upload** (install the plugin if missing).
2. If there is no **data/** folder in the sketch directory, create one — it can be empty. This ensures a clean SPIFFS partition.
3. Upload the filesystem image. This wipes any previous WiFi credentials or fid_map from the device.

[WARNING]

Always format SPIFFS before flashing a device destined for a new deployment. Old credentials or fingerprint mappings from a previous use will cause unexpected behaviour.

3.4 Flash the Firmware

1. Select the correct board: **Tools** → **Board** → **ESP32 Dev Module** (or your specific variant).
2. Select the correct port under **Tools** → **Port**.
3. Click **Upload**. Wait for "Done uploading".
4. Open Serial Monitor at **115200 baud**. The boot sequence should end with:

```
No WiFi credentials found. Launching captive portal...
```

4. First-Boot Verification

Work through this section with the device connected to a laptop via USB and Serial Monitor open at 115200 baud.

1. Power on the device. LED should pulse purple. Serial Monitor shows: *No WiFi credentials found. Launching captive portal...*
2. On your phone, connect to WiFi network **Attendance-Setup** (password: setup1234).
3. Navigate to **http://192.168.4.1**. The WiFi setup form should appear.
4. Enter the site WiFi SSID and password. Click **Save & Connect**.
5. Device restarts. LED should turn solid blue. Serial Monitor shows the IP address, *RTC synced from NTP, Fingerprint sensor found*, and *All tasks started*.
6. If the LED stays purple after restart, the WiFi password was wrong or the network is out of range. Repeat steps 2–4 to re-enter credentials.

[TIP]

Take a photo of the Serial Monitor output during this step as a record that the device was working correctly at setup time.

5. Enrolling the Master Finger

The master finger is used by the administrator to trigger the WiFi captive portal when credentials need to change. It never records attendance.

Add a row to the Enrollment sheet with the following values:

Column	Value
FingerID	Leave blank
UniqueID	MASTER-01
Name	Admin Master (or the administrator's name)
Role	master
Command	register
Status	Leave blank
Note	Leave blank

1. Add the row to the Enrollment sheet. The device polls every 10 seconds.
2. Within 10 seconds, the LED turns solid yellow. Place the administrator's finger on the sensor within 20 seconds.
3. Lift when the LED goes off. Place the same finger again when the LED returns to yellow.
4. LED flashes green on success. The Status column updates to **registered** and FingerID is filled in.
5. Verify: scan the master finger. The LED should flash yellow 5 times and the captive portal should launch. Unplug and replug to exit without reconfiguring WiFi.

[CRITICAL]

Do not assign Role = "staff" to the master finger. It would record attendance instead of launching the portal.

6. Enrolling Staff Fingerprints

Enrol all current staff before handover. The administrator can add more later using the same process.

Add a row to the Enrollment sheet for each staff member:

Column	Value
FingerID	Leave blank
UniqueID	Staff ID, e.g. EMP-001. Must be unique per person.
Name	Staff member's full name
Role	staff
Command	register
Status	Leave blank
Note	Leave blank

1. Add the row. The device picks it up within 10 seconds; LED turns yellow.
2. The staff member places their finger on the sensor and holds it still.
3. Lift when the LED goes off. Place the same finger again when yellow returns.
4. LED flashes green. Status updates to **registered**; FingerID is filled in.
5. Ask the staff member to do a test scan immediately. A row should appear in the Attendance sheet within 30 seconds.
6. Repeat for each staff member.

[NOTE]

The device can store up to **127 fingerprint templates** (slots 1–127). Slot assignment is automatic. A scan cooldown of 60 seconds is enforced per finger — a second scan within that window is silently ignored. After 60 seconds the same finger can scan again and the Time-Out column will be updated.

7. End-to-End Functional Test

Run all of these checks with the device fully set up and connected to the site WiFi.

#	Test	Pass Condition
1	Attendance scan	Scan an enrolled finger. A row appears in the Attendance sheet (Time-In filled) within 30 seconds.
2	Duplicate block	Scan the same finger again within 60 seconds. No new row is added; no change to the existing row.
3	Time-Out update	Wait 60+ seconds and scan again. The existing row's Time-Out column is updated.
4	Late status	Scan an enrolled finger with a timestamp more than 10 minutes past 7:30 AM. Status reads Late.
5	Unrecognised finger	Scan an unenrolled finger. LED flashes red; no sheet entry is created.
6	Master finger	Scan the master finger. LED flashes yellow 5 times; captive portal launches. Unplug to exit.
7	Delete enrollment	Add a delete row in Enrollment for a staff member (using their UniqueID). After 10s, Status changes to deleted. That finger no longer registers a scan.
8	Offline queue	Disable WiFi, scan an enrolled finger, then restore WiFi. The queued scan appears in the sheet within 60 seconds of reconnection.
9	Nightly job (manual)	Run <code>runNightlyJob()</code> from the Apps Script editor. Staff with a Time-In but no Time-Out get No Check-Out; staff with no scan get an Absent row.

8. Handover Checklist

Go through every item below before handover. Do not hand over a device with any unchecked items.

Apps Script

- SCHOOL_SPREADSHEET_ID is correctly set
- SCHOOL_NAME is set to "D and D Academy"
- SHIFT_START_HOUR/MIN = 7/30 and SHIFT_END_HOUR/MIN = 15/30
- LATE_GRACE_MINUTES and EARLY_GRACE_MINUTES = 10
- Admin email addresses are added to adminEmails array
- Script is deployed as Web App (Anyone, Execute as Me)
- SCRIPT_URL has been confirmed live with a test GET request in a browser
- runNightlyJob trigger is set for 10pm–11pm daily
- sendMonthlySummary trigger is set for the 1st of each month, 1am–2am

Firmware

- SCRIPT_URL in firmware matches the deployed Apps Script URL
- SCHOOL_ID = "DDA" and SCHOOL_NAME = "D and D Academy"
- SCRIPT_AUTH matches SCRIPT_AUTH_KEY in the Apps Script (or both are empty)
- SPIFFS was formatted clean before flashing
- Device boots without errors (confirmed via Serial Monitor)

Device

- Connected to site WiFi — LED is steady blue
- RTC time is correct — visible as "RTC synced from NTP" in Serial Monitor on boot
- Master finger is enrolled and verified (portal launches on scan)
- All current staff have been enrolled
- Each enrolled staff member has done a test scan (row appeared in Attendance sheet)
- Attendance sheet is visible, correctly formatted, and accessible to the administrator
- Monthly summary tested (or sendMonthlySummary run manually and output checked)

Handover

- Administrator has the school spreadsheet URL bookmarked or shared with them
- Administrator has been shown how to enrol a new staff member
- Administrator has been shown how to remove a staff member (delete command in Enrollment sheet)
- Administrator has been shown how to use the master finger to reconfigure WiFi
- Administrator has been given (or will receive) the Administrator Manual
- Your personal admin email address has been removed from adminEmails, or the school's address has been added

[TIP]

Take a photo of the Serial Monitor output during the final boot as a record that the device was working correctly at handover.